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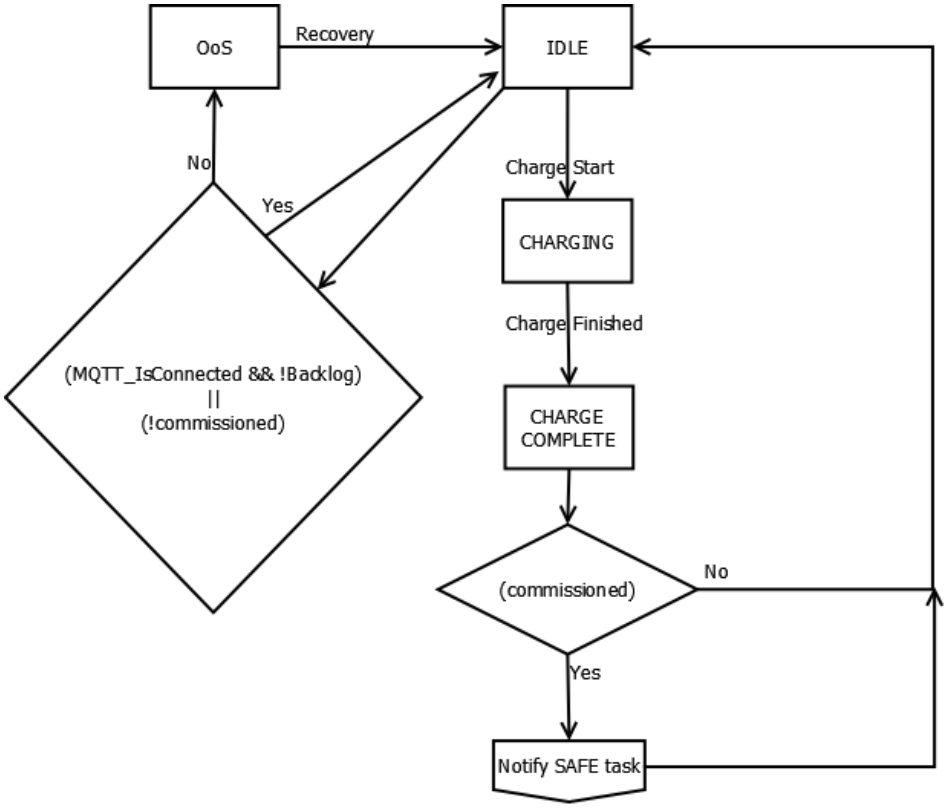
Charge Session State Machine

Overview

The Charge Session State Machine (CSSM) is executed periodically (approximately every 50ms). All states communicate to the AFE measurement module to extract measurement data.

It moves between several major states:

1. Idle State
    a. Monitors voltage/current for the start of a charge session
    b. When start of charging has been detected transitions to Charging State
2. Charging State
    a. Calculates the accumulating charge session energy in kWh
    b. Calculates timing for energy IR pulses
    c. Updates the charge session receipt
    d. Tests for the end of a charge session
    e. When end of charging has been detected transitions to Charge Completion State
3. Charge Completion State
    a. Completes the charge session receipt
    b. Notifies other relevant modules of completion of charge session
    c. Transitions to Idle State
4. Out Of Service (OoS) State
    a. Any state can transition to OoS on occurrence of an unexpected error, either from the CSSM or prompted by supervisor
    b. The CSSM will enter OoS if a previous SAFE packet wasn't successfully sent out over MQTT (Backlog flag shown in diagram)
    c. When allowed by supervisor, monitors conditions for recovery back to Idle - to recover all errors must be cleared and the voltage and current must be below the start thresholds



Tariffs

The DC Meter supports a tariff name to be displayed locally on the Trusted Screen Overlay and logged with the charge session information (as the OCMF optional "UT" field in the first record in the signed transaction data). No fiscal calculations are supported.

The DC Meter is configured in production with a default tariff name that will be used on power up.

The CPO can choose to update the active tariff throughout the meter's life through OCPP. Alternatively the Tariffs can be left with the default name.

Tariff name changes are signed with the CPO's private key. The signature is verified with the CPO public key which is pre-written into the meter's memory also during production. The CPO public key will conform to SECP256R1 (Key Curve), ASN.1 DER (Format), RFC-5480 (Standard).

Once tariffs are sent to the meter they will be used until new tariffs are sent or a power cycle. The tariff for a given charge session will be set once power delivery has started and changing the active tariff after this time will only apply to the next charge session.

Session Energy Calculations

Measured Energy Values (Ws)

Measured energy values of a CSSM calculation interval are calculated using 16 digit fixed point decimal numbers with a range of 10^6 to 10^-9,

These are the values that are reported during each cycle of the CSSM. Energy values are handled to .000000001 Ws precision.

Cumulative Energy Values (kWh)

Cumulative Energy Values are calculated using 19 digit fixed point decimal numbers with a range of 10^6 to 10^-12. Cumulative Energy values are handled to .000000000001 kWh precision.

Error Handling

In the case of an error in energy calculations the CSSM is placed in OoS state.

Accuracy Compensation

To improve the metrological accuracy several methods are applied, see Metrological Accuracy Compensation.

Out Of Service Causes

The following is a list of errors which can cause the CSSM to go out of service

- INVALID_STATE
- START_END_THRESHOLD
- ABOVE_IDLE_ENTRY_THRESHOLD
- MONOTONIC_TIMESTAMP
- AFE_READ
- AFE_CABLE_SOURCE
- GET_MEASUREMENT
- DECNUM_READ_VOLTAGE
- DECNUM_READ_CURRENT
- DECNUM_COMP_VOLTAGE
- DECNUM_COMP_CURRENT
- DECNUM_COMP_CABLE_SOURCE
- DECNUM_SESSION_ENERGY
- DECNUM_PULSE_CALCULATE
- DECNUM_TME
- CABLE_SELECT_SOURCE
- RTC_READ
- RD_CSS_DETAILS
- WR_CSS_DETAILS
- FORCED_CHARGE_COMPLETE
- MQTT_DISCONNECTED
- MQTT_BACKLOG
- TSO_NO_SESSION
- SUPERVISOR
    - Supervisor can put CSSM into OoS for several reasons:
        - Self checks fail (see Self Tests)
        - Firmware update in progress
        - PCBA or Sub-assembly test running

Analogue Front End (AFE)

Overview

The AFE is a physical modular component dedicated for measuring DC voltage and current as accurately as possible. The measurements are then passed to the Charge Session State Machine to manage the high level tasks related to charging.

The AFE is responsible for:

1. performing the Voltage and Current measurements of a charging session so power delivery can be accurately calculated
2. performing regular drift calibration of the ADC
3. performing regular temperature measurements of the ADC
4. automatically handling of ADC gain adjustments for different measured current values
5. performing 'dithering' on the request for measured values to provide a level of immunisation against harmonics
6. converting each measured sample to an actual nV and nA, TempC with a monotonic timestamp
7. providing an API allowing retrieval of measured samples
8. closed analogue loop control of the heating element to maintain 70 degrees Celsius

The Charge Session State Machine is responsible for:

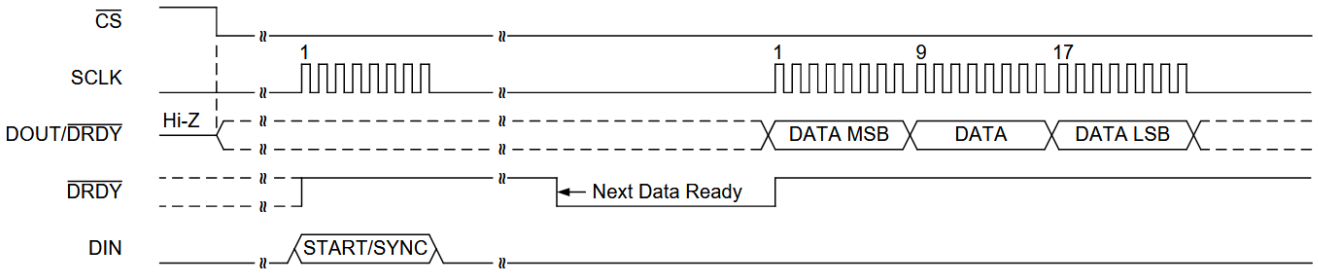
1. calculating kWh from the voltage and current measurements supplied by the AFE
2. accumulating the kWh value for a charge session
3. accumulating the kWh for the life of the meter
4. attribute kWh values to specific tariffs
5. pairing charge session id from the charger with the charge session
6. notify firmware module to sending completed charge session details to the CPO
7. notify firmware module to store the charge session details to persistent storage
8. determining when a charge session starts and ends

AFE Sequencing

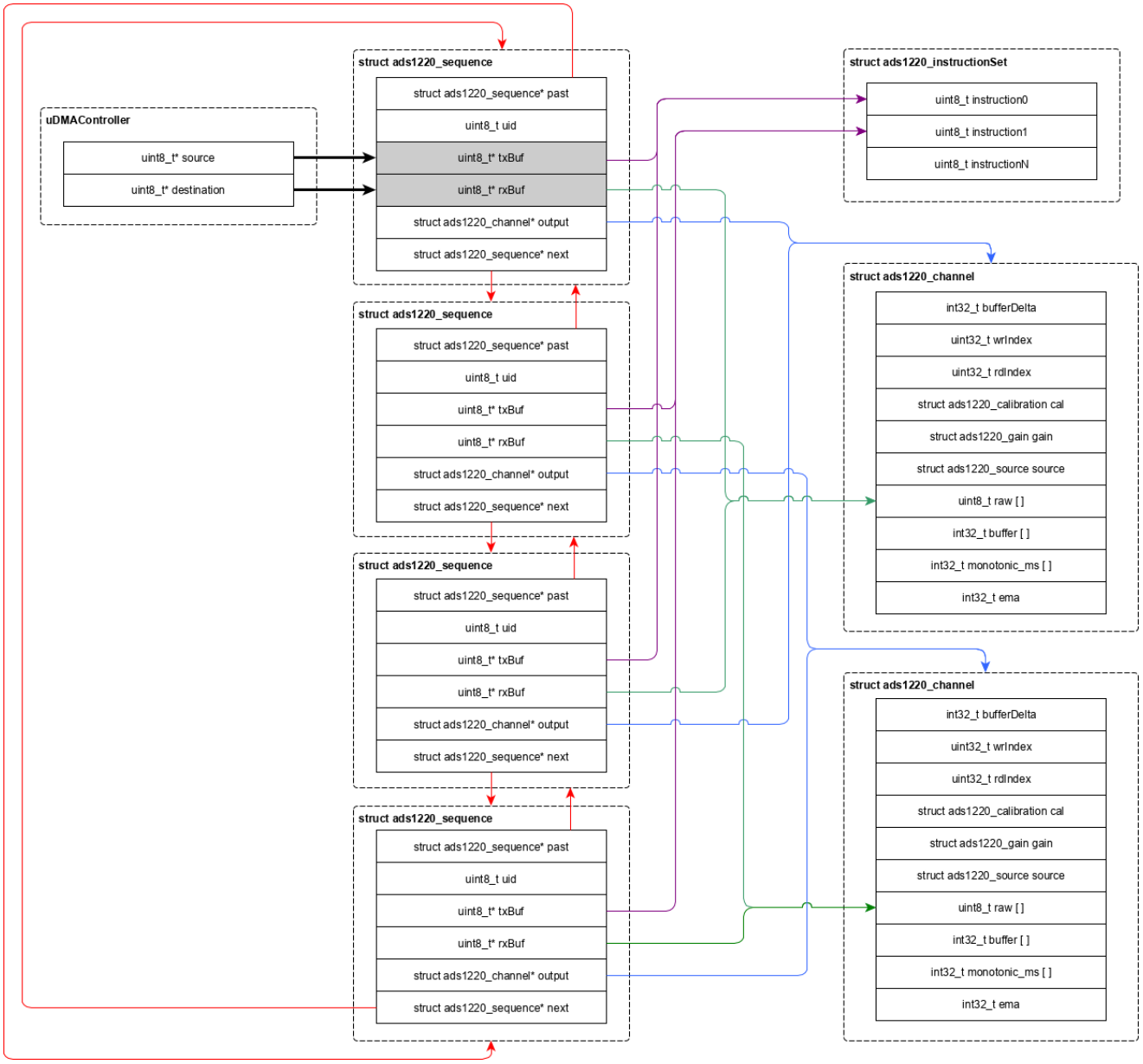
The DC Meter AFE utilises an external 4 channel 24-bit delta sigma ADC (ADS1220).

http://www.ti.com/product/ADS1220

The ADS1220 is configured to operate in continuous single-shot mode. As soon as the **Data-Ready** line of the ADS1220 is low, a ADS1220 instruction set is written to the bus. The data returned on the bus will not correspond to the present ADS1220 instruction. The data returned will be from the previous ADS1220 instruction.



To streamline communication to the ADS1220, the microcontroller's uDMA controller is used. At runtime a dynamically linked list of ADS1220 sequence is created. Every time data is available, the uDMA controller traverses and executes to the next ADS1220 instruction in the list.


uDMAController
uint8_t* source
uint8_t* destination
struct ads1220_sequence
struct ads1220_sequence* past
uint8_t uid
uint8_t* txBuf
uint8_t* rxBuf
struct ads1220_channel* output
struct ads1220_sequence* next
struct ads1220_sequence
struct ads1220_sequence* past
uint8_t uid
uint8_t* txBuf
uint8_t* rxBuf
struct ads1220_channel* output
struct ads1220_sequence* next
struct ads1220_sequence
struct ads1220_sequence* past
uint8_t uid
uint8_t* txBuf
uint8_t* rxBuf
struct ads1220_channel* output
struct ads1220_sequence* next
struct ads1220_sequence
struct ads1220_sequence* past
uint8_t uid
uint8_t* txBuf
uint8_t* rxBuf
struct ads1220_channel* output
struct ads1220_sequence* next
struct ads1220_instructionSet
uint8_t instruction0
uint8_t instruction1
uint8_t instructionN
struct ads1220_channel
int32_t bufferDelta
uint32_t wrIndex
uint32_t rdIndex
struct ads1220_calibration cal
struct ads1220_gain gain
struct ads1220_source source
uint8_t raw [ ]
int32_t buffer [ ]
int32_t monotonic_ms [ ]
int32_t ema
struct ads1220_channel
int32_t bufferDelta
uint32_t wrIndex
uint32_t rdIndex
struct ads1220_calibration cal
struct ads1220_gain gain
struct ads1220_source source
uint8_t raw [ ]
int32_t buffer [ ]
int32_t monotonic_ms [ ]
int32_t ema

LMT01 Pulse Count Temperature Sensor

Datasheet

http://www.ti.com/lit/ds/symlink/lmt01.pdf

Overview

The LMT01 device is a high-accuracy, 2-pin temperature sensor with an easy-to-use pulse count current loop interface, which makes it suitable for on-board and off-board applications in automotive, industrial, and consumer markets. The LMT01 digital pulse count output and high accuracy over a wide temperature range allow pairing with any MCU without concern for integrated ADC quality or availability, while minimising software overhead.

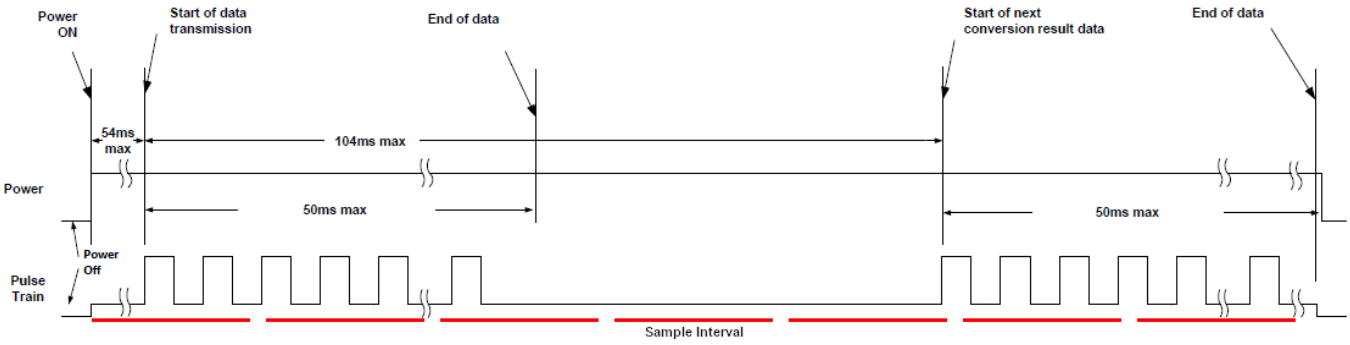
TI's LMT01 device achieves a maximum ±0.5°C accuracy (using non-linear transfer function) with very fine resolution (0.0625°C) over a temperature range of –20°C to 90°C without system calibration or hardware and software compensation. Overall valid measurement range is -50/150degC.

Firmware Considerations

The LMT01 is used for measuring the shunt temperature of the DC Meter. This is considered a safety critical measurement and therefore it's performance and reliability should be ensured.

The main difficulty arises from the fact that the LMT01 device operates asynchronous to the microcontroller.

Since the sensor operates asynchronously, pulse counts are sampled over fixed intervals. The interval time guarantees that one interval will not detect any pulses, signifying the end of a result transmission.



Hardware Considerations

- The LMT01 sensor pulse output to the microcontroller is inverted due to the transistor arrangement that the sensor drives.
  - Therefore the output is inverted relative to the figure shown above.
- The ISO7763 digital isolator holds its output high when the input side is not powered, or the input pin is floating.
- The LMT01 is continuously powered by the AFE supply and therefore its output cannot be easily synchronised to the microcontroller.

Metrological Accuracy Compensation

The DC Meter has a number of mechanisms to improve the accuracy of the measurements. This document describes the overall measurement method and major compensation elements. There is no long term compensation mechanism.

Hardware

The electronic hardware design has been designed to be robust and reliable. One major element implemented in hardware to improve accuracy is the use of an ovenised circuit section which maintains 70°C to minimise thermal drift.

Firmware

Note: Any value with a superscript X (e.g. A^X) is changed dependant on the detected charge cable. Any value with a superscript Y (e.g. A^Y) is changed dependant on the current gain selected.

Energy Calculation Overview

Energy is calculated during charging as a sum using the corrected voltage and current at a fixed interval.

$$E_{TOT} = \sum E_{INT}$$

$$E_{INT} = V_C^X * I_C * t_{INT}$$

Where:

E_{TOT} is total energy during a charge (kWh).

E_{INT} is the energy in the calculated interval (kWh).

V_C^X is the compensated voltage measurement (V).

I_C is the compensated current measurement (A).

t_{INT} is the fixed time interval between calculations (seconds).

Compensated Voltage

Voltage is compensated from the raw samples as follows.

$$V_C^X = [(V_F^X - O_F) * G_V - V_{OFF}^X] * C_V^X - I_C * R_{CABLE}^X$$

Where:

V_F^X is the filtered reading of voltage (ADC Counts).

O_F is the filtered reading of offset of the ADC with 1x gain (ADC Counts)

G_V is the fixed gain to convert to voltage (V/ADC Counts).

V_{OFF}^X is the calibrated offset voltage (V).

C_V^X is the calibrated scale for voltage (V/V).

R_{CABLE}^X is the loaded parameter for resistance of the detected cable (Ohms). Default value is 0.0.

The cable voltage drop based on current will be done within the firmware module doing the energy calculation.

Compensated Current

Current is compensated from the raw samples as follows.

$$I_C = [(I_F^Y - O_F^Y) * G_I^Y - I_{OFF}^Y] * C_I^Y$$

Where:

I_F^Y is the filtered reading of current (ADC Counts).

O_F^Y is the filtered reading of offset of the ADC (ADC Counts).

G_I^Y is the fixed overall gain to convert to current (A/ADC Counts).

I_{OFF}^Y is the calibrated offset current (A).

C_I^Y is the calibrated scale for current (A/A).

Current Sampling Gain

Current is sampled with a different gain depending on the last readings. When the current is rising above 150A the ADC gain is switched to low gain (x64). When the current is falling below 100A the ADC gain is switched to high gain (x128). This allows improved resolution at the lower currents.

Offset Sampling

The offset is obtained by changing the ADC multiplexer to short both positive and negative inputs pins and reading the result. The offset is read with the three different gain levels used, x1, x64, x128 and stored in different registers.

Optical Test Output

To act as an indicator for energy measured, an infrared LED is used as an optical output for certification testing purposes.

This output has two main functions:

- To indicate total measured energy through total number of pulses
- To indicate rate of energy delivery via time between pulses (to be compatible with existing industry standard test equipment)

As this output is used to validate the metrological accuracy, timing inconsistencies cannot be added as a result of execution delays.

Parameters

Constant: 50 Wh/impulse; 20 impulses/ kWh

Delay: 245 milliseconds

Pulse Width: Individual impulse width is 250 microseconds

Operating Points

Maximum Expected Power

At 720 kW, above the maximum power output 350kW of the charger, this will result in 4 pulses per second.

Minimum Expected Power

At 5A; 50V; 250W, the minimum accuracy operation conditions of the DC Meter, this will result in a pulse every 720 seconds.

Implementation

Energy calculations are carried out during a session by the CSSM. Nominally this is done every 50ms, with some variance. The energy calculations are performed based on the AFE samples acquired since the last execution period, the samples are taken nominally at 40ms, with some variance.

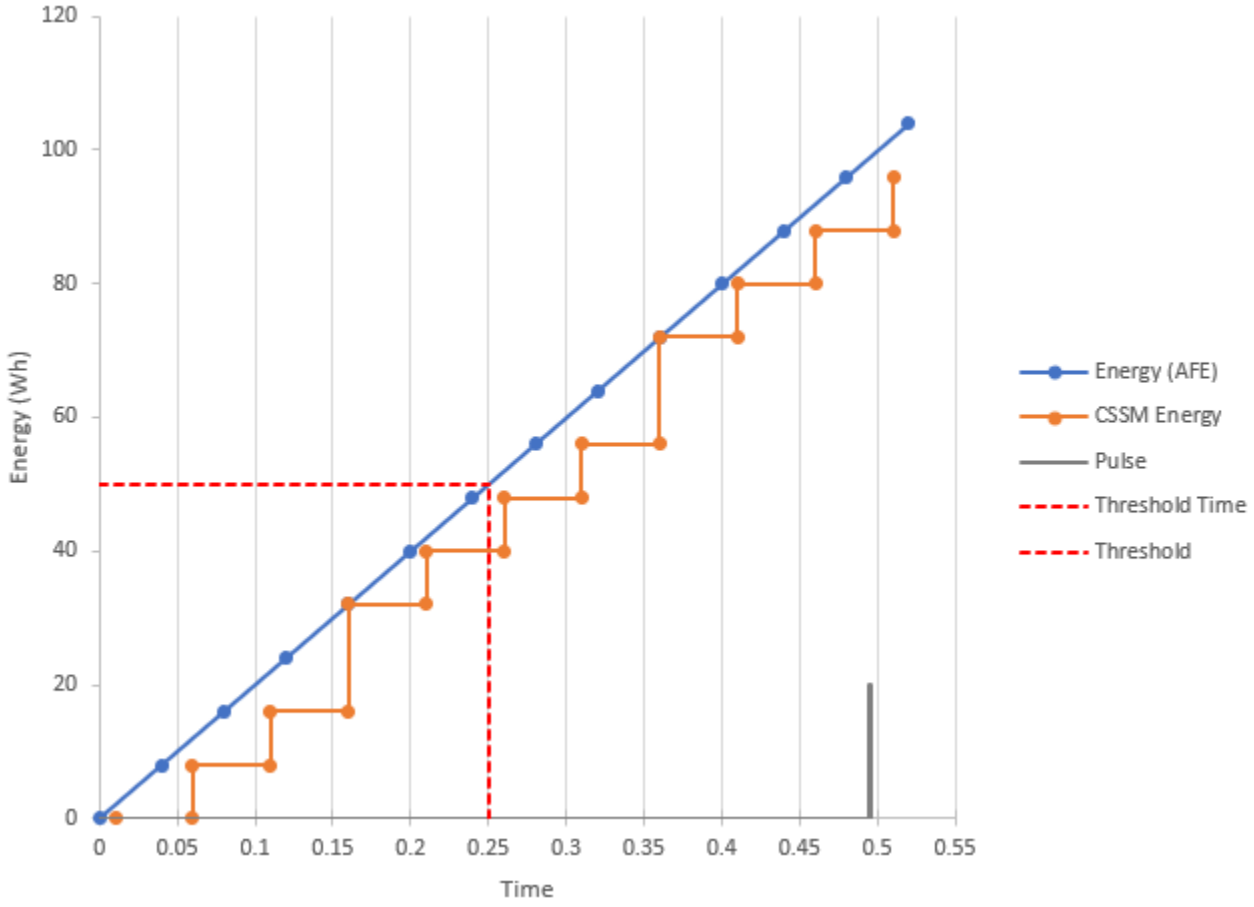
The implementation takes the timing variances into account and produces evenly spaced pulses when at a fixed power level.

Each AFE sample consists of a voltage and current measurement pair that are time-stamped when received by the processor. During the CSSM execution, if the accumulating energy since the last pulse goes above the constant threshold the next pulse is calculated. The time when that threshold was met is calculated by linearly interpolating the energy over the last set of sample measurements and timestamps. This calculation may occur up to ~85ms after the threshold was actually met. A pulse is then scheduled for exactly 245ms after the calculated time.

Peripherals

Three peripherals are used to implement this solution.

- 32 bit timer - used to give accurate timestamps to the measurement interval (with microsecond resolution)
- 32 bit timer - used to precisely schedule the pulse start time
- 16 bit PWM - used to provide a single shot pulse, to drive the hardware pin controlling the infrared LED output


120
100
80
60
40
20
0
Energy (Wh)
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
0.55
Time
Energy (AFE)
CSSM Energy
Pulse
Threshold Time
Threshold